

Page 1 — Introduction to Artificial Intelligence

Artificial Intelligence

Artificial Intelligence (AI) is the field of building computer systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing patterns, solving problems, learning from examples, and making decisions.

Why AI is useful

AI can process large amounts of information quickly and can automate repetitive tasks. Modern AI applications are used in customer support, recommendation systems, document analysis, software development, education, and many other areas.

Machine Learning

Machine Learning (ML) is a subfield of AI. Instead of writing every rule manually, developers provide data and an algorithm learns patterns from that data. Supervised learning uses labeled examples, unsupervised learning searches for patterns without labels, and reinforcement learning learns through interaction and feedback.

Test fact

The fictional company used in this document is called Northstar Labs. Its internal AI assistant is named Nova.

Page 2 — Large Language Models

Large Language Models

A Large Language Model (LLM) is a machine learning model designed to work with human language. LLMs can generate text, summarize information, answer questions, transform text, and help with programming tasks.

Tokens

LLMs process text as tokens rather than simply reading complete words. A token can represent a word, part of a word, punctuation, or another piece of text. Token limits determine how much information a model can process in one request.

Prompt and response

A prompt is the input provided to a model. The model analyzes the prompt and produces a response based on its learned patterns and the information available in the request.

Important limitation

An LLM can produce an answer that sounds confident even when the information is incorrect or missing. This is one reason applications often need grounding mechanisms such as retrieval.

Test facts

Nova uses an LLM to generate natural-language responses. The model is not the same thing as the document database.

Page 3 — Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with a language model. Instead of asking the language model to answer only from its internal training, a RAG application first searches a knowledge source for relevant information.

Basic RAG flow

The typical flow is: user question → retrieval → relevant document chunks → prompt with retrieved context → language model → grounded answer.

Documents and chunks

Large documents are usually divided into smaller chunks before indexing. Chunking makes it easier for a retriever to find the specific section related to a question. Chunk size and overlap can affect retrieval quality.

Embeddings

An embedding is a numerical representation of text. Text with similar meaning can have similar vector representations. A vector store can use these representations to search for semantically related content.

Test fact

In the fictional Nova system, each document is split into chunks before embeddings are created. The chunks are then stored in a vector store.

Page 4 — Retrievers and Vector Stores

Vector Store

A vector store is a system that stores vector representations and associated information. During a RAG query, the application can compare the query representation with stored vectors and retrieve relevant chunks.

Retriever

A retriever is the component responsible for finding relevant information from a knowledge source. It receives a query and returns documents or chunks that may help answer the question.

Similarity search

Similarity search compares the representation of a query with stored representations. Depending on the implementation, different distance or similarity measures can be used to identify relevant results.

Metadata

Metadata is additional information associated with a document or chunk. Examples include the source filename, page number, document category, author, or creation date. Metadata can be useful for filtering and displaying citations.

Test facts

Nova's vector store contains 120 document chunks. The system stores the original source filename as metadata. The retriever normally returns the top 4 relevant chunks.

Page 5 — LangChain RAG Application

LangChain and RAG

LangChain provides components that can be used to build applications around language models. A RAG application can connect document loaders, text splitters, embedding models, vector stores, retrievers, prompts, and chat models into an application workflow.

Production considerations

A production RAG system should consider retrieval quality, chunking strategy, prompt design, latency, cost, access control, document freshness, logging, evaluation, and protection against prompt injection.

Evaluation

RAG systems should be tested with representative questions. Useful evaluation areas include whether the correct documents are retrieved, whether the answer is supported by the retrieved context, and whether the system correctly refuses questions that are outside its knowledge source.

Final test facts

The fictional project is called Smart Knowledge Assistant. Its assistant is Nova. It uses 120 document chunks, retrieves 4 chunks per query, and stores source filenames as metadata. The company name is Northstar Labs.

Suggested RAG questions

1. What is the assistant's name? 2. What company created it? 3. How many document chunks are in the vector store? 4. How many chunks are retrieved per query? 5. What metadata is stored? 6. What is RAG? 7. Why are documents split into chunks?